**Example 11.17**

Integrate the following with respect to $x$:

(i) $\dfrac{e^{2x}-1}{e^x}$ (ii) $e^{3x}\left(e^{2x}-1\right)$.

**Solution**

(i) $$\int \frac{e^{2x}-1}{e^x}dx = \int\left(\frac{e^{2x}}{e^x}-\frac{1}{e^x}\right)dx$$

$$= \int\left(e^x-e^{-x}\right)dx = e^x+e^{-x}+c.$$

(ii) $$\int e^{3x}\left(e^{2x}-1\right)dx = \int\left(e^{5x}-e^{3x}\right)dx = \frac{e^{5x}}{5}-\frac{e^{3x}}{3}+c.$$

**Example 11.18**

Evaluate : $\int \dfrac{1}{\sin^2 x\cos^2 x}dx.$

**Solution**

$$\int \frac{1}{\sin^2 x\cos^2 x}dx = \int \frac{\sin^2 x+\cos^2 x}{\sin^2 x\cos^2 x}dx$$

$$= \int \frac{1}{\cos^2 x}dx+\int \frac{1}{\sin^2 x}dx$$

$$= \int \sec^2 x dx+\int \operatorname{cosec}^2 x dx$$

$$= \tan x-\cot x+c.$$

**Example 11.19**

Evaluate : $\int \dfrac{\sin x}{1+\sin x}dx.$

**Solution**

$$\int \frac{\sin x}{1+\sin x}dx = \int\left(\frac{\sin x}{1+\sin x}\right)\left(\frac{1-\sin x}{1-\sin x}\right)dx$$

$$= \int \frac{\sin x-\sin^2 x}{1-\sin^2 x}dx = \int \frac{\sin x-\sin^2 x}{\cos^2 x}dx = \int \frac{\sin x}{\cos^2 x}dx-\int \frac{\sin^2 x}{\cos^2 x}dx$$

$$= \int \tan x\sec x dx-\int \tan^2 x dx$$

$$= \int \tan x\sec x dx-\int (\sec^2 x-1)dx$$

$$= \sec x-\tan x+x+c.$$

**Example 11.20**

Evaluate : $\int \sqrt{1+\cos 2x}\,dx.$

**Solution**

$$\int \sqrt{1+\cos 2x}\,dx = \int \sqrt{2\cos^2 x}\,dx = \sqrt{2}\int \cos x dx = \sqrt{2}\sin x+c$$



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